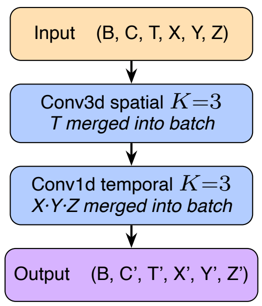
Patchify architecture — PatchEmbed3DPlus1D

2026-06-14

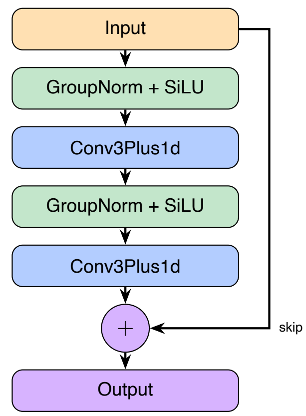
Hierarchical 3D+1D patchify encoder, inspired by the MovieGen TAE `TemporalEncoder` (tae.py:1052).

Building blocks

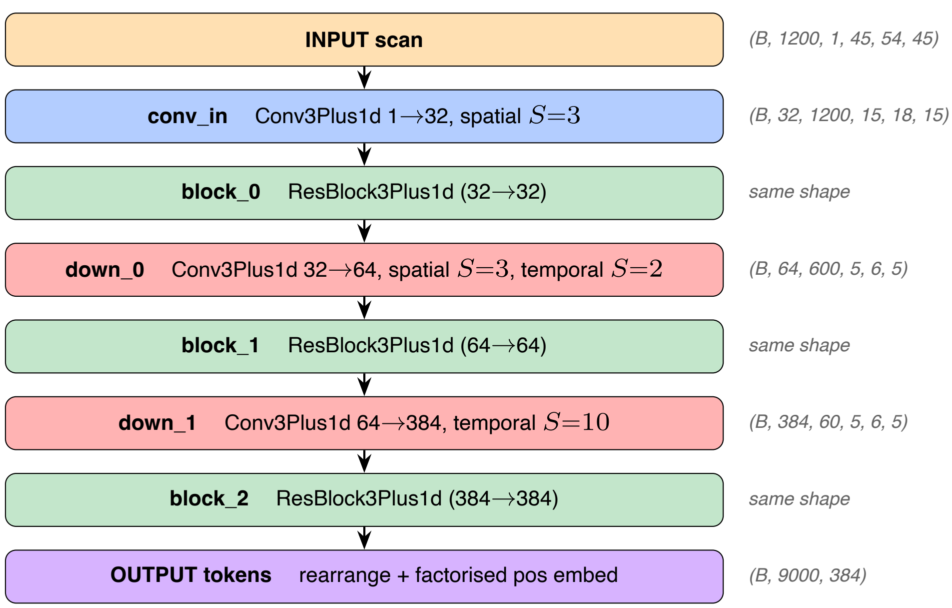
Conv3Plus1d — separable substitute for the missing PyTorch `Conv4d`. Splits a 4D conv into a 3D spatial conv (per frame) followed by a 1D temporal conv (per voxel-position).



ResBlock3Plus1d — residual block of 2 `Conv3Plus1d` with GroupNorm + SiLU pre-activation, output added to a skip connection. Same shape in / out.



Full encoder (HCP-YA setup, T = 1200, temporal_kernel = 20)



Parameter breakdown

Layer	What it contains	Params
`conv_in`	1 Conv3Plus1d (1{→}32)	4 000
`block_0`	2 Conv3Plus1d (32{→}32) + 2 GroupNorm	61 696
`down_0`	1 Conv3Plus1d (32{→}64)	67 712
`block_1`	2 Conv3Plus1d (64{→}64) + 2 GroupNorm	246 272
`down_1`	1 Conv3Plus1d (64{→}384)	2 138 880
`block_2`	2 Conv3Plus1d (384{→}384) + 2 GroupNorm	8 850 432
`PositionEmbedding3D`	`pos_temporal` + `pos_spatial` + `pos_cls`	81 024
TOTAL		**11 450 016**

Token grid output : $T_{\text{eff}} = 60$, $N_{\text{spatial}} = 5 \cdot 6 \cdot 5 = 150$, total **9 000 tokens** of dim 384.

Differences vs MovieGen TAE `TemporalEncoder`

Aspect	MovieGen TAE	Ours (PatchEmbed3DPlus1D)
Domain	Video (T, H, W)	fMRI volumes (T, X, Y, Z)
Building block	`Conv2Plus1d` (2D spatial + 1D temporal)	`Conv3Plus1d` (3D spatial + 1D temporal)
Hierarchical levels	4 (default `ch_mult=(1,2,4,8)`)	3 (channels 32 / 64 / 384)
ResBlocks per level	2	1
Attention in encoder	Yes (at specified resolutions)	None
Middle bottleneck section	2 ResBlocks + 1 AttnBlock	None
Output	Latent tensor `z` (for decoder)	Token sequence (B, $T_{\text{eff}} \cdot N_{\text{spatial}}$, 384) for ViT